

## Quiz - Monod's Nightmare

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My name is: \_\_\_\_\_

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Under ideal growing conditions in the laboratory, the population size of the common bacterium *Escherichia coli* can double in around 20 minutes. *E. coli*'s cells are rod-shaped and are roughly approximated by a rectangular box that is  $2\ \mu m$  long,  $1\ \mu m$  wide, and  $1\ \mu m$  high. ( $1\ \mu m = 1\ \text{micro-meter} = 1 \times 10^{-6}\ \text{meters} = 0.000001\ \text{m}$ ). For comparison, a strand of human hair is roughly  $100\ \mu m$  wide.

(a) Would a continuous-time model (the solution for which is  $N_t = N_0 e^{rt}$ ) or a discrete-time model (the solution for which is  $N_t = \lambda^t N_0$ ) be more appropriate for describing this population? Why?

Continuous. Reproduction occurs continuously, not in discrete synchronous bouts.

(b) What is *E. coli*'s growth rate  $r$  (in minutes) under these ideal growing conditions. Express your answer by solving for the equation you would use to calculate  $r$  to the simplest solution possible.

$$\begin{aligned} t &= 20\ \text{minutes}; \quad N_t = 2N_0 \\ N_t &= N_0 e^{rt} \\ \ln\left(\frac{N_t}{N_0}\right) &= rt \\ r &= \frac{\ln\left(\frac{N_t}{N_0}\right)}{t} = \frac{\ln(2)}{20} \approx \frac{0.693}{20} \approx 0.035\% \text{ per minute} \end{aligned}$$

(c) Let's assume that our classroom is roughly  $10\ m$  long,  $10\ m$  wide, and  $8\ m$  high (it's not, I just made up these numbers). How long it would take for an exponentially growing population of *E. coli* under ideal conditions to fill our empty classroom when starting from a single individual bacterium?

The volume of each *E. coli* is

$$\begin{aligned} 1\ \mu m \cdot 1\ \mu m \cdot 2\ \mu m &= 2\ \mu m^3 \\ 1 \cdot 10^{-6}\ m \cdot 1 \cdot 10^{-6}\ m \cdot 2 \cdot 10^{-6}\ m &= 2 \cdot 10^{-18}\ m^3 \end{aligned}$$

The volume of room is  $10\ m \cdot 10\ m \cdot 8\ m = 800\ m^3$

Therefore, the number of *E. coli* needed to fill room is

$$\frac{800\ m^3}{2 \cdot 10^{-18}\ m^3} \approx 4 \cdot 10^{20}.$$

Thus:

$$\begin{aligned} N_0 &= 1 \\ N_t &= N_0 e^{rt} \\ \ln\left(\frac{N_t}{N_0}\right) &= rt \\ t &= \frac{\ln\left(\frac{4 \cdot 10^{20}}{1}\right)}{r} = \frac{\ln\left(\frac{4 \cdot 10^{20}}{1}\right)}{\frac{\ln(2)}{20}} \approx \frac{47.438}{0.035} \approx 1369\ \text{minutes} \approx 22.8\ \text{hours} \end{aligned}$$